

EXP NO: 5

CLUSTERING WITH K-MEANS AND DIMENSIONALITY REDUCTION WITH PCA

AIM

To implement K-Means clustering for grouping similar data points and Principal Component Analysis (PCA) for dimensionality reduction and visualization using a suitable dataset.

PROCEDURE

1. Import the required Python libraries.
2. Generate a suitable dataset using `make_blobs()`.
3. Apply the Elbow Method to determine the optimal number of clusters.
4. Apply K-Means clustering with the selected number of clusters.
5. Visualize the clusters and their centroids.
6. Calculate the Silhouette Score to evaluate the quality of clustering.
7. Generate a higher-dimensional dataset for PCA.
8. Standardize the dataset using `StandardScaler`.
9. Apply PCA to reduce the dimensionality of the dataset from 4 dimensions to 2 dimensions.
10. Calculate the explained variance ratio of the principal components.
11. Visualize the PCA-transformed data.
12. Apply K-Means clustering to the PCA-reduced data.
13. Visualize the clusters obtained from the PCA-reduced data.
14. Calculate the Silhouette Score for the PCA-based K-Means clustering.

CODE

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.datasets import make_blobs

from sklearn.preprocessing import StandardScaler

from sklearn.decomposition import PCA
```

```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

X, y = make_blobs(
    n_samples=500,
    n_features=4,
    centers=4,
    cluster_std=1.0,
    random_state=25
)

df = pd.DataFrame(
    X,
    columns=["Feature_1", "Feature_2", "Feature_3", "Feature_4"]
)

print(df.head())
print("Original Shape:", df.shape)

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
```

```
X_pca = pca.fit_transform(X_scaled)

df_pca = pd.DataFrame(
    X_pca,
    columns=["PC1", "PC2"]
)

df_pca["True_Cluster"] = y

print("\nPCA Data:")
print(df_pca.head())

variance = pca.explained_variance_ratio_

print("\nExplained Variance Ratio:", variance)

print(
    "Total Explained Variance:",
    round(variance.sum(), 3)
)

plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(  
    x="PC1",  
    y="PC2",  
    hue="True_Cluster",  
    data=df_pca,  
    palette="Paired",  
    s=100  
)  
  
plt.title("PCA - Dimensionality Reduction")  
plt.xlabel("Principal Component 1")  
plt.ylabel("Principal Component 2")  
plt.grid()  
plt.show()  
  
kmeans = KMeans(  
    n_clusters=4,  
    init="k-means++",  
    max_iter=300,  
    n_init=10,  
    random_state=42  
)  
  
clusters = kmeans.fit_predict(X_pca)
```

```
df_pca["KMeans_Cluster"] = clusters
```

```
plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(  
    x="PC1",  
    y="PC2",  
    hue="KMeans_Cluster",  
    data=df_pca,  
    palette="viridis",  
    s=100  
)
```

```
plt.scatter(  
    kmeans.cluster_centers_[:, 0],  
    kmeans.cluster_centers_[:, 1],  
    s=300,  
    marker="X",  
    label="Centroids"  
)
```

```
plt.title("K-Means on PCA-Reduced Data")
```

```
plt.xlabel("Principal Component 1")
```

```
plt.ylabel("Principal Component 2")
```

```
plt.legend()
```

```
plt.grid()
```

```
plt.show()
```

```
score = silhouette_score(X_pca, clusters)
```

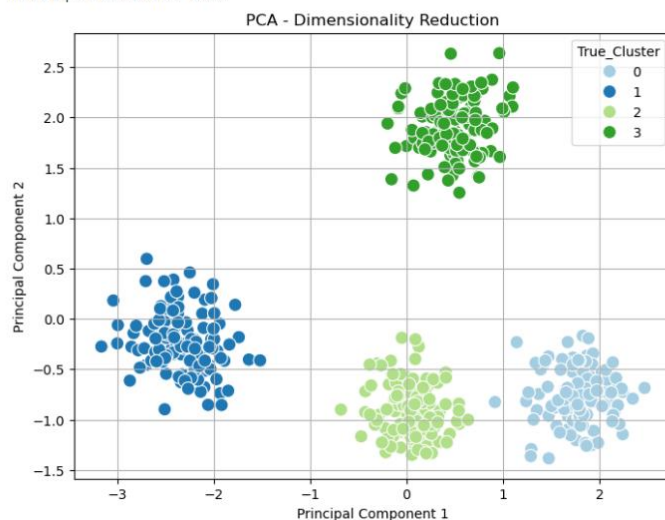
```
print("Silhouette Score:", round(score, 3))
```

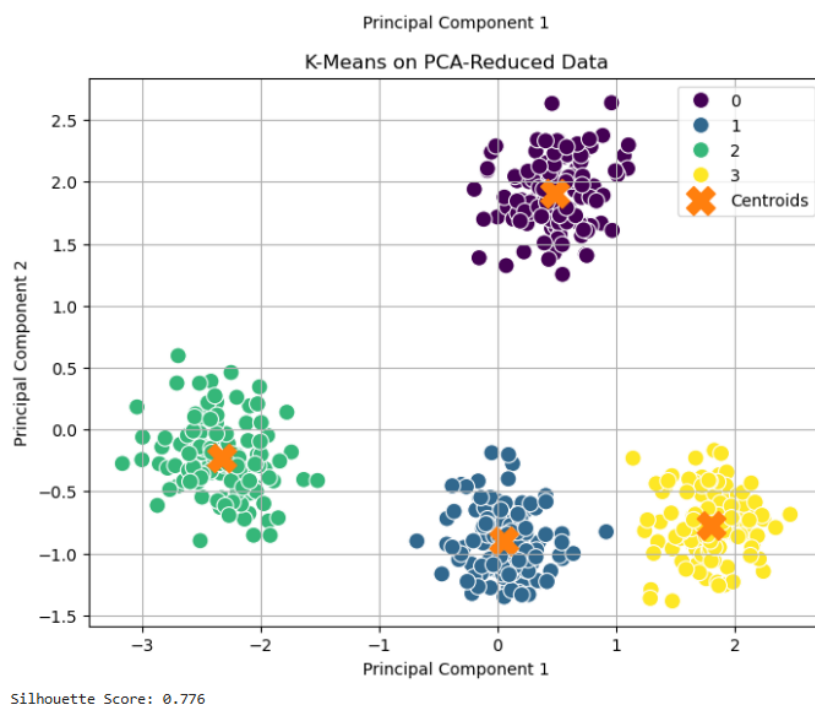
OUTPUT

```
Feature_1 Feature_2 Feature_3 Feature_4
0 -0.638667 1.110057 -6.400722 -0.204990
1 -2.951556 -7.657445 3.844794 0.903589
2 -0.253177 2.125103 -7.869801 0.559678
3 -2.151209 3.401400 -5.734930 0.965230
4 -2.347519 -7.230467 3.478891 -0.443440
Original Shape: (500, 4)
```

```
PCA Data:
PC1 PC2 True_Cluster
0 0.455305 1.623917 3
1 -2.705622 0.375012 1
2 0.810234 1.966926 3
3 0.427139 2.149626 3
4 -2.407508 0.099250 1
```

```
Explained Variance Ratio: [0.57208431 0.33622342]
Total Explained Variance: 0.908
```





RESULT

PCA was successfully applied to the four-dimensional dataset and reduced it to two principal components while retaining the major variance in the data. K-Means clustering was then performed on the PCA-reduced data, and the resulting clusters were visualized. The Silhouette Score was calculated to evaluate the clustering quality. Thus, PCA effectively simplified the high-dimensional data and K-Means successfully identified the underlying clusters.